

DESIGN PHASE

1.Introduction

The Design Phase is a critical stage in the development of the Laptop Request Management System. In this phase, the overall structure, workflow, and functionality of the system are planned and designed before implementation. The main objective of this phase is to create an automated process that simplifies the handling of laptop requests within an organization.

The system is developed using the ServiceNow platform and utilizes various modules such as Service Catalog, Flow Designer, Approval Management, Catalog Tasks, and Email Notifications. The design ensures that every laptop request follows a standardized workflow from submission to fulfillment.

2.System Architecture Design

The Laptop Request Management System follows a workflow-based architecture. The process begins when an employee submits a laptop request through the Service Catalog. The request is then routed to the designated approver for review. Based on the approval decision, the system automatically performs the next actions such as creating a task for the IT team and sending notifications to the requester.

The architecture consists of the following components:

Employee/User:

The employee acts as the requester who submits the laptop request through the Service Portal.

Service Catalog:

The Service Catalog provides a user-friendly interface for employees to submit laptop requests. It captures the necessary information required for processing the request.

Flow Designer:

Flow Designer serves as the automation engine of the system. It controls the workflow and performs actions based on predefined conditions.

Approval Engine:

The Approval Engine manages the approval process by sending approval requests to managers or authorized approvers.

Catalog Task:

Catalog Tasks are automatically generated after approval and assigned to the IT support team for fulfillment.

Notification System:

The notification system keeps users informed about the status of their requests through email alerts.

3.Workflow Design

The workflow has been designed to automate the complete request lifecycle. Once the user submits a laptop request, the system automatically initiates the approval process. The request remains in a pending state until the approver makes a decision.

If the request is approved, a Catalog Task is created for the IT team, and an approval notification is sent to the employee. If the request is rejected, a rejection email is automatically sent to the requester.

The workflow helps reduce manual effort and ensures that requests are processed efficiently and consistently.

4.User Interface Design

The user interface is designed using ServiceNow Service Catalog variables. The form is simple and easy to use, allowing employees to submit requests without technical expertise.

The Laptop Request form includes the following fields:

- Employee Name
- Department
- Laptop Type
- Justification
- Requested For

The design focuses on simplicity, usability, and data accuracy.

5.Approval Process Design

The approval process is designed to ensure that all laptop requests are reviewed before resources are allocated. When a request is submitted, the system automatically sends an approval request to the designated approver.

The approver can either approve or reject the request based on business requirements. This approval mechanism provides better control over organizational assets and prevents unauthorized requests.

6.Task Management Design

After approval, the system automatically generates a Catalog Task for the IT support team. The task contains all necessary details required to fulfill the request.

The task management process ensures:

- Proper assignment of responsibilities.
- Tracking of task progress.
- Timely delivery of laptops.
- Improved accountability.

This automation reduces delays and eliminates manual task creation.

7.Notification Design

The notification module is designed to improve communication between the system and users. Email notifications are triggered automatically based on the request status.

Approval Notification

When a request is approved, the employee receives a confirmation email informing them that the IT team will process the request.

Rejection Notification

If the request is rejected, the employee receives an email explaining that the request was not approved and may contact the manager for additional information.

This feature improves transparency and keeps users informed throughout the process.

8.Data Flow Design

The system follows a structured data flow where information moves through different stages of the workflow. The request data is collected through the Service Catalog, processed by Flow Designer, validated through the Approval Engine, and finally used to generate tasks and notifications.

This design ensures accurate processing, data consistency, and effective tracking of requests.

9.Security Considerations

The system incorporates role-based access control to ensure that only authorized users can access specific functions. Employees can submit requests, managers can approve requests, and IT personnel can fulfill tasks.

Security measures implemented include:

- User authentication
- Role-based permissions
- Approval validation
- Activity tracking and auditing

These controls help maintain data integrity and system reliability.

10.Design Outcome

The designed Laptop Request Management System provides a fully automated and efficient process for handling laptop requests. The system minimizes manual effort, improves approval efficiency, enhances communication through automated notifications, and ensures proper tracking of requests.

The design is scalable, user-friendly, and capable of supporting future enhancements such as asset management integration, AI-based recommendations, and automated procurement processes. The successful completion of the design phase establishes a strong foundation for the implementation and deployment of the system within the ServiceNow platform.